

Evaluation Criteria — L1- inception-stories-generator- agent

Quality Gates (must pass)

Criterion	Threshold	Method
All features decomposed	100% coverage	Automated: count features in input vs features with stories
Story points ≤ 5	No violations	Automated: check every story
AC count 3-5 per story	No violations	Automated: count per story
Given/When/Then format	100% AC	Automated: regex check
Dependencies acyclic	No cycles	Automated: graph traversal
No intra-feature Tier 1 deps	Foundation independent within feature	Automated: dependency check
Title format	"As a / I want / So that"	Automated: regex check
All stories have implements[]	No empty	Automated: field check

Evaluation Scores (LLM-as-Judge)

Evaluator	Threshold	Direction
Faithfulness	≥ 0.90	Every story traces to a feature/FR
Hallucination	≤ 0.10	No invented capabilities beyond input
Consistency	≥ 0.90	Stories don't contradict each other
Relevance	≥ 0.85	Stories are implementable, not vague
Reasoning quality	≥ 0.80	Tier/points/sequence decisions explained
Citation completeness	≥ 0.95	Every story cites feature_id and FR

Story Quality Rubric

Dimension	Score 9-10	Score 7-8	Score 5-6	Score < 5
Decomposition	Feature fully decomposed, right-sized stories	Most stories right-sized, one may be too large	Some stories too large or too small	Stories don't decompose the feature

Dimension	Score 9-10	Score 7-8	Score 5-6	Score < 5
Sequencing	Perfect tier ordering, team can implement in sequence	Mostly correct, one minor ordering issue	Several sequencing issues	No logical order
AC quality	All testable, Given/When/Then, specific outcomes	Most testable, minor vagueness	Several vague AC	AC untestable
Dependencies	Acyclic, minimal, enable parallel work	Correct but could be more parallel	Some unnecessary deps	Circular or missing deps
Completeness	All FRs + NFRs addressed	FRs covered, NFRs partially	Some FRs missing	Major gaps

Reflection Checklist

The agent must self-verify before delivering:

- ☐ Every feature has ≥ 2 stories
- ☐ Tier 1 stories exist for every feature (setup/foundation)
- ☐ No story exceeds 5 points
- ☐ Dependencies allow a team to start immediately (Tier 1 has no intra-feature deps)
- ☐ NFRs from epic are reflected in at least one story (usually Tier 3)
- ☐ Acceptance criteria are specific enough for a developer to implement and QA to test
- ☐ Execution summary includes recommended cycle sequence

Reflection Process (mandatory)

The agent **MUST** perform reflection before delivering output:

1. **Generate** initial output following processing rules
2. **Log** [REFLECTING] Checking output against evaluation criteria
3. **Check** every item in the Reflection Checklist above
4. **Identify** gaps, errors, inconsistencies, or missed items
5. **Log** each finding: [REFLECTING] Found: <description>
6. **Fix** each issue — amend the output silently
7. **Log** each resolution: [REFLECTING] Resolved: <what was fixed>
8. **Deliver** only the final corrected output

The reflection findings and resolutions should appear in the execution_summary (what reflection found and changed) but the interim output must never be shown.